

LESSON 3-4 • SESSION 1 • PRACTICE • FORM B

Graphing Ratios on the Coordinate Plane

Learning target – I can turn a ratio into ordered pairs and plot them on a coordinate plane. · 6.RP.A.3a

REMEMBER

- An **ordered pair** (x, y) names one point. Go **across** x first, then **up** y.
- Each column of a ratio table is one point. Points from equivalent ratios form a **straight line** through **(0, 0)**.
- To test a point, multiply its x by the rate. Do you get its y?

WORKED EXAMPLE

Each tray holds 8 muffins (x = trays, y = muffins). Write and plot three points.

Step 1 Table: 1 tray = 8, 2 trays = 16, 3 trays = 24.

Step 2 Pairs: (1, 8), (2, 16), (3, 24).

Step 3 Plot each pair from (0, 0): across x, then up y.

Answer The points lie on one straight line through (0, 0)

WORDS coordinate plane (*plano de coordenadas*) · ordered pair (*par ordenado*) · origin (*origen*)

PRACTICE

show your work · label units

1. Each bag holds 8 soccer balls (x = bags, y = balls). Write the ordered pair for:

a. 1 bag _____

b. 3 bags _____

c. 5 bags _____

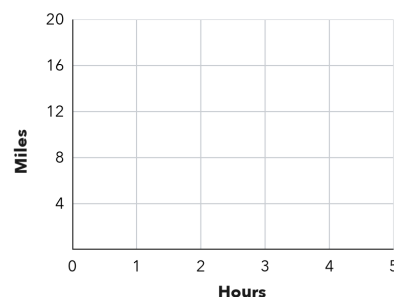
d. 7 bags _____

2. **Hiking Trail.** Hikers climb 4 miles every hour (x = hours, y = miles).

a. Complete: (1, 4), (2, ____), (3, ____), (4, ____)

b. Plot the points. Draw a line through (0, 0).

c. What point shows 7 hours? _____



3. **File Downloads.** A computer downloads 3 GB every 2 minutes (x = minutes, y = GB).

Hint 10 minutes is 5 groups of 2 minutes.

a. Complete: (2, 3), (4, 6), (6, ____)

b. GB in 10 minutes: _____ GB

Show your work.

4. **Street Trees.** A crew plants 9 trees on every block (x = blocks, y = trees).

a. Complete: (1, 9), (2, ____), (3, ____)

b. Point for 5 blocks: _____

Show your work.

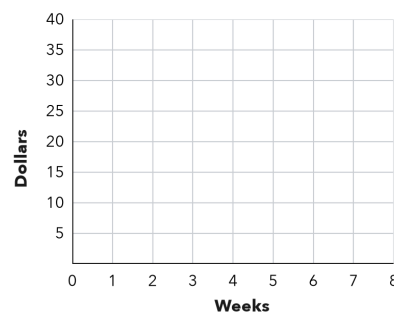
5. **Allowance Savings.** Diego saves \$5 each week (x = weeks, y = dollars).

Hint Multiply each number of weeks by 5.

a. Complete: (1, 5), (2, ____), (4, ____), (6, ____)

b. Find y for the point (7, y). _____ dollars

c. Plot your points and label the point for 7 weeks.



- 6. Poster Printing.** A printer prints 6 posters each minute (x = minutes, y = posters).

Hint The equation is $y = (\text{rate}) \times x$.

a. Write the equation: $y =$ _____

b. What does (0, 0) mean here? _____

- 7. Scooter Range.** A scooter goes 12 miles on each battery charge (x = charges, y = miles). Which point is on its graph: (3, 38), (4, 48) or (5, 55)?

Hint Multiply each x by 12. Does it match y ?

Show your work.

Point _____

- 8. Two Stores.** Store A sells notebooks at 2 for \$5. Store B sells them at 5 for \$15. Graph x = notebooks, y = cost.

a. Cost of 10 notebooks at each store: _____

b. Whose line is steeper? What does steeper mean here? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Mateo's Work.** A train travels 60 miles each hour (x = hours, y = miles). Mateo plotted the points (60, 1) and (120, 2). Explain his mistake and write the correct points.

TEST PRACTICE

choose the best answer

- 10.** In the ordered pair (7, 21), the number 21 is the –

A x-value

B y-value

C unit rate

D origin

- 11.** The ratio of x to y is 1 : 4. Which ordered pair has $x = 5$?

A (5, 9)

B (20, 5)

C (5, 20)

D (4, 5)

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

(4, 48) S	(7, 28) O	15 I	(7, 56) P	(56, 7) Q	(28, 7) X	(5, 45) N	(48, 4) Z	35 T
#1d	#2c	#3b	#4b	#5b	#7			

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____